Findings & Model Evaluation

Stepwise Logistic Regression Models

To evaluate the performance of the logistic regression models, three complementary metrics were used:

- Area Under the ROC Curve (AUC)
- McFadden's Pseudo R^2
- Brier Score

Each metric evaluates a different aspect of model performance.

1. Area Under the ROC Curve (AUC)

What does AUC measure?

The Area Under the Receiver Operating Characteristic (ROC) Curve measures the model's ability to discriminate between active drug users and former users.

Rather than relying on a single classification threshold (e.g., 0.5), AUC evaluates the model's performance across all possible thresholds. It can be interpreted as the probability that a randomly selected active user will receive a higher predicted probability than a randomly selected former user.

Interpretation Scale

AUC	Interpretation
0.50	Random classification
0.60–0.70	Weak
0.70–0.80	Fair to Good
0.80–0.90	Strong
>0.90	Excellent

2. McFadden's Pseudo R^2

What does McFadden's R^2 measure?

Because logistic regression does not provide the traditional R^2 used in linear regression, McFadden's Pseudo R^2 is commonly used as an alternative measure of model fit.

It compares the final model with a null model containing only an intercept and evaluates how much explanatory power is gained by including the predictor variables.

Interpretation Scale

McFadden's R^2	Interpretation
<0.05	Weak
0.05–0.10	Moderate
0.10–0.20	Good
>0.20	Strong

Unlike linear regression, values between 0.05 and 0.10 are generally considered acceptable for logistic regression models.

3. Brier Score

What does the Brier Score measure?

The Brier Score evaluates the accuracy of the predicted probabilities generated by the model.

It is calculated as the mean squared difference between the predicted probabilities and the observed outcomes. Therefore, it reflects how closely the predicted probabilities match the actual observations.

Unlike AUC and McFadden's R^2, **lower values indicate better performance**.

Interpretation

- 0 indicates perfect probability predictions.
- Lower values represent more accurate probability estimates.
- Higher values indicate poorer calibration of predicted probabilities.

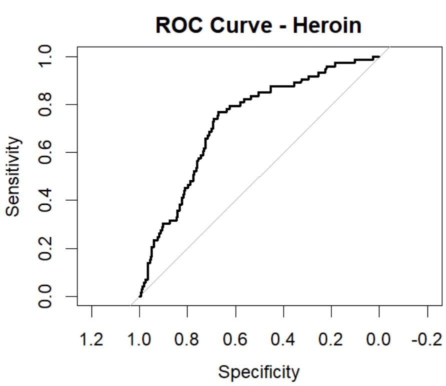
Expanding Sample Model

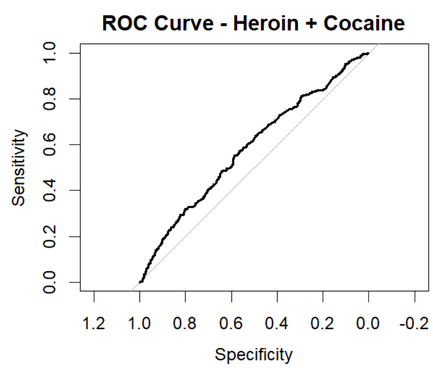
Performance Evaluation of the Expanding Sample Models

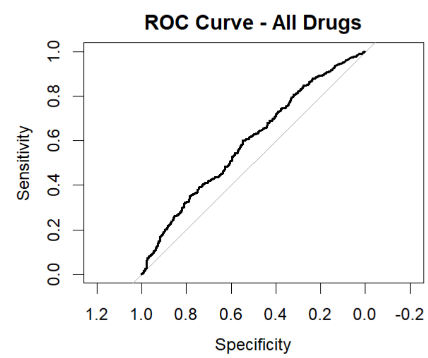
As an initial modeling strategy, a single stepwise logistic regression framework was applied to progressively larger study populations. The analysis began with individuals who had a history of heroin use, followed by a model including both heroin and cocaine users, and finally a model including heroin, cocaine, and methamphetamine users. This approach was designed to evaluate whether the same socioeconomic and mental health predictors remained informative as the study population expanded and became more diverse. By progressively increasing the sample size, we aimed to assess the generalizability and stability of the selected predictors across broader groups of substance users.

Although several predictors remained statistically significant throughout the expanding models, the overall model performance gradually declined as additional drug types were combined. Specifically, the AUC and McFadden's Pseudo R^2 decreased, while the Brier Score increased, suggesting reduced discriminative ability and explanatory power as the study population became more heterogeneous. The performance of each model was evaluated using AUC, McFadden's Pseudo R^2, and Brier Score, as presented in Table.

	Model	AUC	McFadden_R2	Brier_Score
1	Heroin	0.7308810	0.08254347	0.0556184
2	Heroin + Cocaine	0.5837709	0.01548568	0.2022440
3	All Drugs	0.5935863	0.01934098	0.2426878







The performance evaluation of the expanding sample models indicated a noticeable decline in model quality as additional drug types were combined into a single analysis. Both the AUC and McFadden's Pseudo R^2 decreased, while the Brier Score increased, suggesting that the models became less effective as the study population became more heterogeneous.

<u>These findings suggested that the relationships between the selected socioeconomic and mental health factors predictors and continued drug use may differ across substances.</u> Therefore, separate stepwise logistic regression models were developed for heroin, cocaine, and methamphetamine users to better capture substance-specific patterns and allow for a direct comparison of the predictors associated with continued use.

Drug-Specific Stepwise Logistic Regression Models

Separate stepwise logistic regression models were developed for heroin, cocaine, and methamphetamine users. This approach enabled the identification of substance-specific predictors of continued drug use and allowed the performance of each model to be evaluated independently.

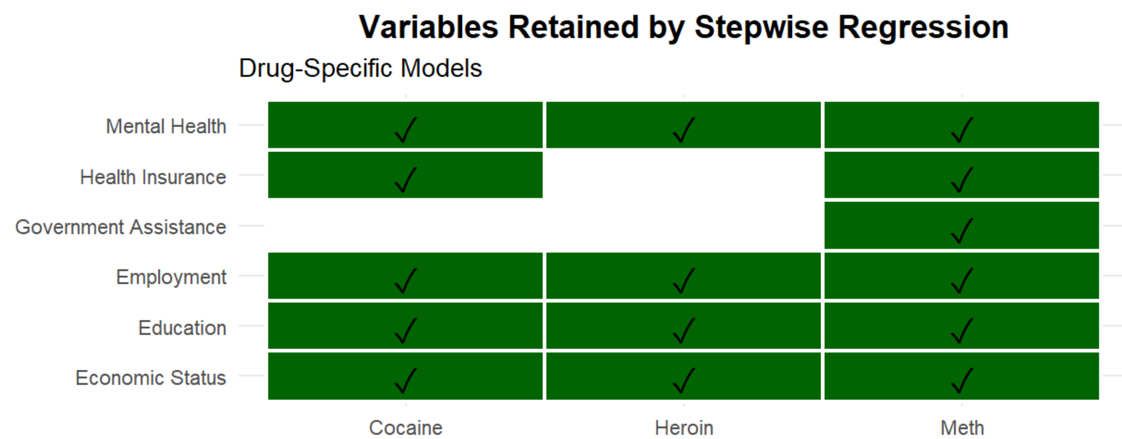


Figure1 presents the variables retained in the final stepwise logistic regression models for heroin, cocaine, and methamphetamine users. **Education, Economic Status, Employment, and Mental Health** were retained in all three models, suggesting that they are common predictors of continued drug use. In contrast, **Health Insurance** and **Economic Assistance** appeared only in specific models, indicating that their contribution may vary across different substances.

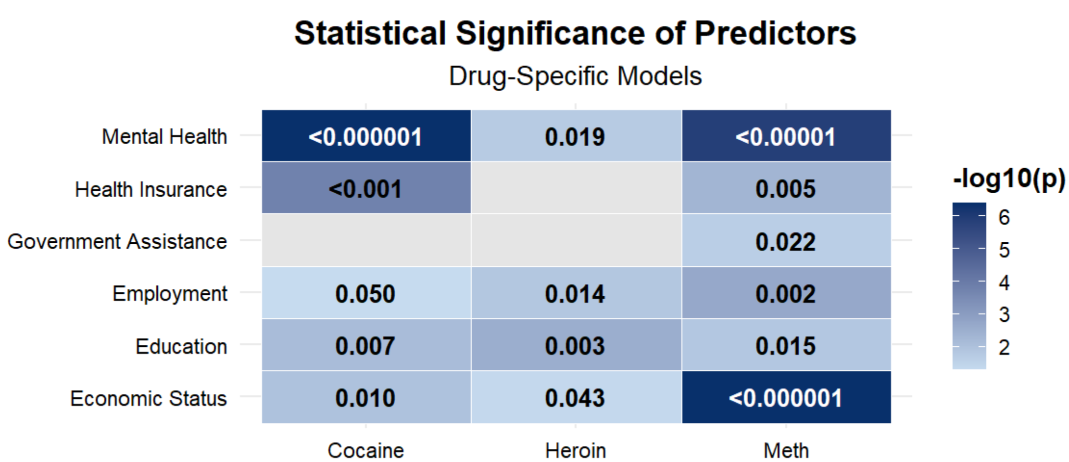
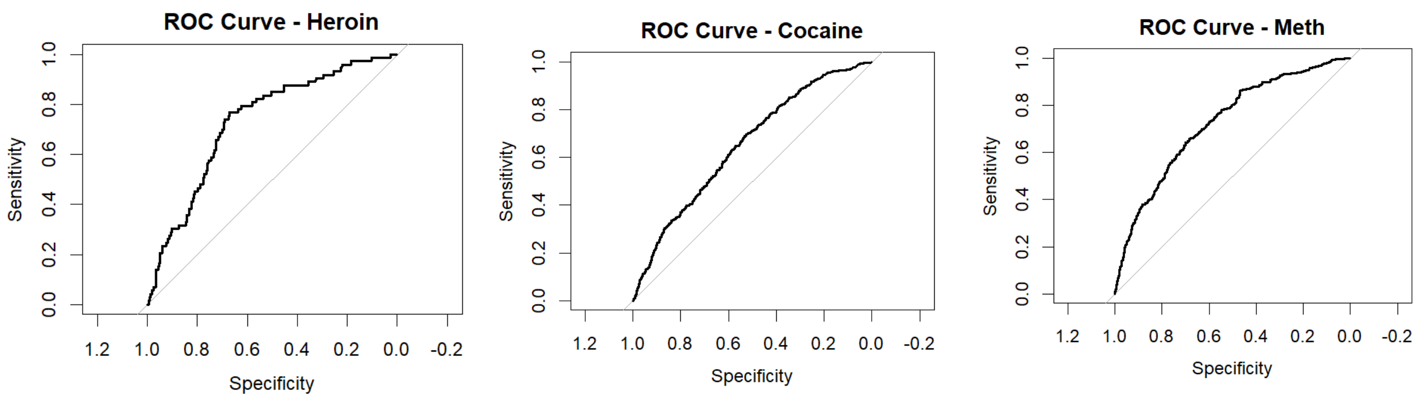


Figure2 presents the statistical significance of the predictors retained in each drug-specific model. **Mental Health** was statistically significant across all three substances and showed particularly strong significance in the cocaine and methamphetamine models. **Education** and **Economic Status** were also significant in all models, supporting their consistent association with continued drug use. **Employment** remained significant for all three substances, although its effect was weaker in the cocaine model. In contrast, **Health Insurance** and **Government Assistance** were significant only in specific drug models, suggesting that their influence may depend on the substance being analyzed.

Model Performance Evaluation

The predictive performance of the drug-specific stepwise logistic regression models was evaluated using three complementary metrics: AUC, McFadden's Pseudo R^2, and Brier Score. Together, these measures assess the models' discriminative ability, explanatory power, and the accuracy of their predicted probabilities. The results are summarized in Table.

	Model	AUC	McFadden_R2	Brier_Score
1	Heroin	0.7308810	0.08254347	0.05561840
2	Cocaine	0.6491038	0.03128309	0.04831225
3	Methamphetamine	0.7255391	0.09646376	0.09854556



Overall, the heroin and methamphetamine models demonstrated the strongest predictive performance. Both achieved **AUC** values of approximately **0.73**, indicating a fair-to-good ability to distinguish between active and former users. In comparison, the cocaine model achieved a lower AUC (**0.649**), suggesting weaker discriminative ability.

A similar trend was observed for **McFadden's Pseudo R^2**. The methamphetamine model achieved the highest explanatory power (**0.096**), followed by the heroin model (**0.083**), whereas the cocaine model showed substantially lower explanatory power (**0.031**). Although these values are relatively modest, they fall within the range commonly reported for logistic regression models and indicate meaningful improvement over the corresponding null models.

The **Brier Score** indicated acceptable probability calibration for all three models. While the cocaine model achieved the lowest Brier Score, this metric should be interpreted cautiously because it is influenced by the prevalence of the positive class and is therefore less appropriate for direct comparisons across datasets with different outcome distributions.

Overall, the evaluation metrics suggest that the selected socioeconomic and mental health variables explain continued heroin and methamphetamine use more effectively than continued cocaine use. These findings further support the use of separate drug-specific models for investigating the factors associated with continued drug use.

Additional hierarchical logistic regression

As part of the initial model development process, a hierarchical logistic regression was also evaluated using the same expanding-sample strategy that was initially explored for the stepwise logistic regression models. Although the final analysis adopted separate drug-specific stepwise logistic regression models, the hierarchical approach produced broadly consistent findings, supporting the main conclusion that socioeconomic and mental health variables remain important factors associated with continued hard-drug use. Therefore, the hierarchical analysis is presented as an additional robustness analysis.

Random Forest Model

To complement this analysis, we employed a Random Forest model. Unlike logistic regression, Random Forest can capture complex nonlinear relationships and interactions between variables without relying on predefined functional assumptions. Furthermore, the model incorporated demographic and geographic variables in addition to the socioeconomic variables, allowing us to evaluate whether the socioeconomic factors remained among the strongest predictors in a broader predictive setting that more closely reflects real-world conditions.

The purpose of the Random Forest analysis was therefore not to replace the statistical interpretation provided by logistic regression, but to provide an additional predictive perspective.

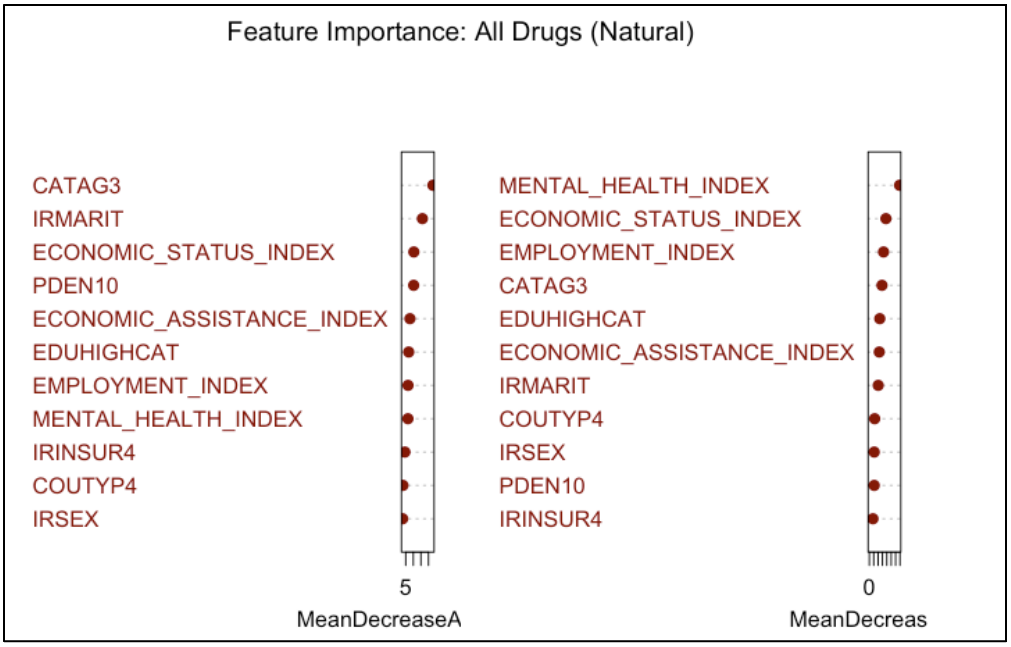
Initial Random Forest Models

Initially, separate Random Forest models were developed for heroin, cocaine, and methamphetamine users in order to examine substance-specific prediction. Although these models demonstrated reasonable discriminatory ability (ROC-AUC approximately 0.70–0.72), they showed relatively limited sensitivity in identifying active users. This is likely explained by the relatively small number of positive cases available for each substance individually, together with the heterogeneity of user characteristics across different drug types.

Since the primary objective of the Random Forest analysis was predictive performance rather than substance-specific interpretation, a broader modeling strategy was subsequently explored.

מדד	הרואין (Heroin)	קוקאין (Cocaine)	מתאמפטמין (Meth)
דיוק (Accuracy)	0.746	0.743	0.756
דיוק מאוזן (Balanced Acc)	0.615	0.629	0.654
רגישות (Sensitivity)	0.466	0.501	0.519
סגוליות (Specificity)	0.764	0.756	0.789
ROC AUC	0.703	0.698	0.724

A final Random Forest model was therefore trained using all hard drug users as a single population. This approach better reflects real world settings, where prevention and intervention programs generally target individuals at risk of continued hard-drug use rather than users of one specific substance. By combining all substances and incorporating demographic, geographic, and socioeconomic variables, we aimed to identify the most influential predictors of continued drug use.



The figure presents the variable importance rankings from the Random Forest model trained on the combined hard-drug dataset. Two complementary importance measures are shown: **Mean Decrease Accuracy**, which reflects the reduction in predictive performance when a variable is removed, and **Mean Decrease Gini**, which reflects each variable's contribution to improving the decision tree splits. Higher rankings indicate variables that provide more information for distinguishing between individuals who continued using hard drugs and those who did not.

The feature importance analysis indicates that socioeconomic variables, particularly **economic status**, **employment**, and **economic assistance**, together with the **Mental Health Index**, remained among the most informative predictors in the Random Forest model. Even after demographic and geographic variables were incorporated, these variables continued to provide substantial information for distinguishing between individuals who continued using hard drugs and those who did not. These findings complement the logistic regression results by demonstrating that socioeconomic variables remain highly informative in a broader predictive setting.

Evaluation of the Combined Random Forest Model

Metric	Value	Interpretation
Accuracy	**0.616**	Overall classification accuracy.
Balanced Accuracy	**0.613**	Average classification performance across both classes.
Sensitivity (Recall)	**0.673**	Ability to correctly identify individuals who continued using hard drugs.
Specificity	**0.552**	Ability to correctly identify individuals who did not continue using hard drugs.
Precision	**0.625**	Proportion of predicted positive cases that were correctly classified.
F1 Score	**0.648**	Harmonic mean of Precision and Recall.
ROC-AUC	**0.654**	Overall discriminative ability of the model.
PR-AUC	**0.673**	Precision–Recall performance, particularly informative for imbalanced datasets.

The combined Random Forest model demonstrated moderate predictive performance. While its overall discriminative ability was moderate (ROC-AUC = 0.654), the model achieved a good balance between sensitivity (0.673) and precision (0.625), resulting in an F1-score of 0.648. These results indicate that the model is reasonably effective in distinguishing between individuals who continued using hard drugs and those who did not.

Among the evaluated Random Forest models, the combined model was selected because it achieved the **highest sensitivity** while maintaining balanced overall performance. Since the primary objective of the Random Forest model was to identify individuals who continued using hard drugs, sensitivity was considered the most important evaluation metric. Therefore, minimizing false negatives was prioritized over maximizing overall classification accuracy.

Random Forest Validation in a Broader Predictive Setting

The final Random Forest model was trained on the combined hard-drug cohort and incorporated socioeconomic, demographic, and geographic variables. Its objective was not to test statistical significance, but rather to examine whether the socioeconomic variables identified by the stepwise logistic regression remained among the most informative predictors after expanding the model to include additional real-world characteristics. As summarized in Table below, the variables consistently retained by the logistic regression models, particularly Mental Health, Economic Status, Employment, and Education, also ranked among the most informative predictors in the Random Forest model. This agreement between the two modeling approaches strengthens the conclusion that these socioeconomic variables remain highly informative for distinguishing between individuals who continued using hard drugs and those who did not, even within a broader predictive framework.

	Variable	Cocaine	Heroin	Meth	RF_Rank
1	Mental Health	✓	✓	✓	1
2	Economic Status	✓	✓	✓	2
3	Employment	✓	✓	✓	3
4	Age				4
5	Education	✓	✓	✓	5
6	Government Assistance			✓	6
7	Marital Status				7
8	County Type				8
9	Sex				9
10	Population Density				10
11	Health Insurance	✓		✓	11